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The cross-spousal spillover effects of social pensions: evidence from time use

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ABSTRACT

This paper examines the cross-spousal spillover effects of social pensions on time use. We exploit the temporal and geographical variation in the county-by-county rollout of China's New Rural Pension Scheme (NRPS) in a difference-in-differences (DID) framework to identify the cross-spousal effects. We find that spousal eligibility for pension benefits significantly changed individuals' time-use patterns, leading to less time spent working and more time spent caring and exercising. These changes can be plausibly attributed to two mechanisms, including the income effect of spousal pension benefits and the optimistic expectation generated by the transparent and predictable cash inflow from spousal pensions in the future. Further investigation shows that spousal eligibility for social pensions can also significantly improve individuals' relationships with non-cohabitant children and enhance their mental health.

KEYWORDS

Cross-spousal; social pensions; time use; China

JEL CLASSIFICATION

H55; J14; I38; O18

I. Introduction

With rapid population ageing and high poverty rates among the elderly, old age support has become a serious social problem worldwide. Many developing countries, such as South Africa (Ardington, Case, and Hosegood 2009), several Latin American countries (Carvalho Filho and Evangelista 2008; Miglino et al. 2023), and India (Kaushal 2014), have responded by introducing or expanding social pension programmes to cover the vast majority of the vulnerable elderly. The public expenditure on these non-contributory pensions is unprecedented. For example, it accounts for at least one percent of the GDP in Peru (Bando, Galiani, and Gertler 2022). Hence, whether it makes sense to spend so much on these pension schemes has been a significant policy issue (Case and Deaton 1998). Meanwhile, the literature has primarily focused on the welfare impact of social pensions on beneficiaries (Bando, Galiani, and Gertler 2022; Kaushal 2014). Although cash transfer programmes might also affect neighbouring ineligible populations (Angelucci and De Giorgi 2009), there is limited knowledge about how social pension programmes for the elderly affect their family members, especially their spouses.

This paper aims to fill the gap by examining the impact of spousal eligibility for pension benefits on one's own time use. We focus on time-use outcomes primarily for two reasons. First, exploring the effects on time use can better reveal an individual's behavioural responses to spousal pensions (Field et al. 2023). Time-use data typically provides information on activity-specific duration, such as time allocated to work, care for families, exercise, and other activities (Chang, Dong, and MacPhail 2011). Hence, our study can comprehensively assess the cross-spousal effects on an individual's behaviours, including labour supply, engagement in home production, and health-improving activities. Second, an individual's time-use response has significant welfare implications. The time-use pattern can directly affect one's subjective well-being and experienced utility (Flores, Ingenhaag, and Maurer 2015). For example, time spent on leisure activities and interacting with families stimulates pleasant feelings, while work-related activities lead to negative feelings (Kahneman et al. 2004). Above all, analysing individuals' time-use responses can reveal the cross-spousal spillover effects of social pensions on their behavioural and well-being outcomes.

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Theoretically, spousal eligibility for pension benefits can affect individuals' time-use patterns primarily through two channels. First, the income effect of spousal pension benefits can change individuals' time-use patterns (Gale 1998; Huang and Zhang 2021). For example, the increased household income from spousal pension benefits might lead to a decline in working hours and an increase in time spent on leisure. Second, spousal pensions can also affect individuals' time allocation by generating optimistic expectations. Specifically, spousal monthly pension benefits are similar to a defined-benefit plan, which indicates transparent and predictable cash inflow in the future. This might reduce household-level uncertainty and change individuals' current time-use patterns (Xu et al. 2022). Meanwhile, spousal pensions may also have minimal effects on individuals' time use primarily for two factors: first, whether social pensions can effectively increase income and create predictable cash inflow at the household level (Ávila-Parra, Escamilla-Guerrero, and Gálvez-Soriano 2024; Hua, Zhang, and Liu 2022), and second, whether the increased income and the cash inflow from spousal pension benefits are shared effectively within couples (Juarez and Pfutze 2015). Above all, further empirical evidence is required to determine whether spousal eligibility for pension benefits affects individuals' time-use patterns.

This paper exploits China's New Rural Pension Scheme (henceforth, the NRPS) to examine the impact of spousal eligibility for pension benefits on individuals' time use. China provides a natural setting for our analysis. First, the rural elderly lacked a universal and generous pension system before the implementation of the NRPS. Second, the NRPS is an unprecedented welfare programme targeting the rural population. It was piloted in 2009 and quickly reached universal coverage by the end of 2012. Once the county was covered, rural elderly aged 60 or older were eligible to receive a nontrivial pension benefit of ¥55 (approximately US $9) per month without contributions.[1] These features inform our sample selection and empirical strategy. Specifically, we primarily focus on the rural populations who themselves are age-ineligible (younger than 60 years old) with spouses age-eligible (aged 60 or older) for pension benefits. Then, we employ a staggered difference-in-differences (DID) framework by exploiting the temporal and geographical variation in the county-level implementation of the NRPS to estimate how individuals' time allocation responds to spousal eligibility for pension benefits. Our empirical strategy requires the common trend assumption, and we provide suggestive evidence supporting this in Section 3.2.

We follow Huang and Zhang (2021) and combine two large nationally representative survey datasets, the China Family Panel Studies (CFPS) and the China Health and Retirement Longitudinal Studies (CHARLS), covering 2010 to 2018, to identify the cross-spousal spillover effects of social pensions. We find that spousal eligibility for the NRPS pension benefits significantly changed individuals' time use. Individuals' weekly time spent working decreased by approximately 5.3 hours, and weekly time spent caring for families and exercising increased by approximately 7.5 and 4.1 hours, respectively. These effects were accrued to the most vulnerable population, women and those from less-developed regions. These results are robust to cascading robustness checks, including controlling for potential confounders, addressing potential bias in staggered DID and data issues from pooling datasets, and placebo tests.

Mechanism analysis suggests that spousal eligibility for pension benefits affected individuals' time use primarily through two channels. First, spousal pension benefits from the NRPS significantly increased household income and thus decreased (increased) individuals' time spent on paid work (leisure activities) through the income effect. Second, the transparent and predictable income from spousal monthly benefits in the future reduced individuals' pessimism and thus decreased (increased) individuals' working hours (time spent caring for families and exercising) through the optimistic expectation effect.

To provide additional evidence on the cross-spousal spillover effects of social pensions, we

[1]This amount of the basic pension benefit is not trivial because there are 5.05 percent of rural elderly with household income per month less than ¥55, and the median monthly income per capita in our analysis data is only ¥302.

further investigate the impact of spousal eligibility on individuals' other socioeconomic outcomes. First, we examine the cross-spousal effect on individuals' relationships with their non-cohabitant children. Results indicate that spousal pensions significantly increased individuals' contact frequency with non-cohabiting children, likely due to public pension benefits crowding out private transfers from children, thereby reducing the financial burden on children. Second, we further examine the cross-spousal effects on individuals' health outcomes. Although we find no discernible impact on physical health, we observe significant improvement in individuals' mental health, potentially due to reduced work time, increased household income, more optimistic expectations, and closer ties with non-cohabitant children.

This paper makes several contributions to the existing literature. First, we contribute to the growing literature on the impact of social pension programmes. Most studies focus on the impact of pension schemes on pension recipients in terms of labour supply (Carvalho Filho and Evangelista 2008), fertility (Danzer and Zyska 2023), health (Miglino et al. 2023), subjective well-being (Bando, Galiani, and Gertler 2022), and a set of welfare indicators such as consumption, income, and assets (Huang and Zhang 2021). Although it has been documented that such cash transfer programmes might affect neighbouring non-eligible people (Angelucci and De Giorgi 2009), we know much less about how extended family members, especially spouses, are affected.[2] The ignorance of such spillover effects might lead to not only identification issues due to the ineligible population not being the ideal counterfactual group (Huang and Zhang 2021) but also the underestimation of social benefits from pension programmes. To the best of our knowledge, this paper is the first to investigate the cross-spousal spillover effects of social pension programmes. Moreover, our findings on significant changes in time-use patterns (less work and more leisure) and improvement in parent-child relationships and mental health provide empirical evidence supporting such social pension programmes.

Second, we also contribute to the literature on the cross-spousal spillover effects. It has been widely documented that couples always make collective decisions such as labour force participation, retirement, and leisure (Chiappori 1992; Georges-Kot, Goux, and Maurin 2024). Individuals' decision-making and welfare can be affected by spousal characteristics and behaviours. For example, spousal education and retirement have nontrivial effects on individuals' health behaviour (Müller and Shaikh 2018) and health outcomes (Guo et al. 2020). Meanwhile, spousal unemployment has significant effects on one's own labour supply (Lundberg 1985), smoking behaviour (Everding and Marcus 2020), and subjective well-being (Nikolova and Ayhan 2019). However, there has been limited exploration of the cross-spousal effects in the social pensions arena. This paper examines how spousal eligibility for pension benefits affects individuals' behaviour and well-being and explores the underlying mechanisms, thus contributing by providing new empirical evidence on the cross-spousal spillover effects.

Third, we also contribute to the understanding of factors influencing time-use patterns. Since Becker's (1965) pioneer theory on the allocation of time, a large literature has documented various determinants of time allocation, including preferences (Stratton 2012), health status (Ozturk and Kose 2019), and minimum wage (Gearhart, Sonchak-Ardan, and Thibault 2023). Moreover, many studies analyse time use within a family (Blundell, Pistaferri, and Saporta-Eksten 2018; Solberg and Wong 1992) and document the role of the birth of sons, caste, the gender earnings gap and the cost of home appliances in the gender disparity in time allocation (Bar and Leukhina 2011; Eswaran, Ramaswami, and Wadhwa 2013; Fan et al. 2018). We contribute by documenting the cross-spousal effects of social pensions on individuals' time use and the potential channels through the income effect and the optimistic expectation effect.

The remainder of the paper is organized as follows. In Section 2, we provide a brief introduction to the NRPS in China. Section 3 describes the data and outlines the empirical specification. We report

[2]Only a few studies have examined the impact of social pensions on extended families (Carvalho Filho and Evangelista 2012; Kaushal 2014).

the main results in Section 4, explore the potential mechanisms in Section 5, and conduct further analysis in Section 6. Finally, Section 7 concludes the paper.

II. Institutional background

In the 1990s, China's rural elderly suffered from poverty and precarious living conditions. China initiated a pension programme to provide economic security for the rural elderly in 1992. However, this Old Rural Pension Scheme was inefficient in providing old-age support to the targeted rural elderly. The failure can be plausibly attributed to the widespread mismanagement of the funds, the insignificance of the programme, and the modest payments afforded by the pension scheme (Huang and Zhang 2021). In 2008, only 55.95 million rural people participated in the scheme, and the enrolment rate was only 5.85 percent. The rural elderly still lacked a universal and generous pension system. In contrast, urban employees had been secured by an urban pension system with high coverage and generous payments since the early 1990s.[3] The surging poverty of the rural elderly and the giant rural-urban inequality in social protection led to significant public concerns.

In reaction to the public infuriation over old-age support in rural areas, the Chinese government launched a large social pension programme, the New Rural Pension Scheme (NRPS), in 2009. This social pension reform had four important features. First, the NRPS was rolled out on a county-by-county basis. The NRPS was initially implemented in 320 pilot counties (approximately 12 percent of all counties) in 2009 and expanded quickly to reach universal coverage in mainland China by the end of 2012 after four rounds of expansions.[4] Second, once a county became covered, all rural people aged 16 or above who were not students and not already covered by an urban pension scheme could voluntarily participate. Third, when the NRPS was first introduced, participants aged 60 or older were eligible to receive a basic pension benefit without paying premiums. The basic transfer is conditioned exclusively on age and irrespective of previous contributions; thus, it is non-contributory and similar to a defined benefit plan that is not means-tested (Huang and Zhang 2021). Moreover, the basic pension benefit of ¥55 (approximately US $9) per month is sufficient to guarantee an elderly person's basic survival in rural China (Huang and Zhang 2021). It is also generous enough given that the median monthly income per capita is only ¥302, with approximately 5.05 percent of households earning less than ¥55 per month in our analysis sample.[5]

The NRPS is an unprecedentedly large and generous social pension programme in rural China that makes substantial progress in social protection for the rural elderly. By the end of 2012, the public fund expenditure on the NRPS totalled more than 262 billion yuan (approximately US$41 billion), with more than 232 billion yuan (about US$37 billion) contributed by the central government. Moreover, as shown in Figure 1, the number of participants and recipients of the NRPS has been increasing since 2009. In 2011, 326 million rural people participated in the NRPS, and 89 million rural pensioners received pension benefits. By the end of 2020, the number of participants had reached 542.44 million, making the NRPS the first such large and universal welfare programme in rural China.

III. Data and empirical strategy

Data

Following Huang and Zhang (2021), we primarily use the data from the China Family Panel Studies (CFPS) and the China Health and Retirement Longitudinal Studies (CHARLS). The CFPS, launched in 2010, biennially collects a nationally

[3]China's social pension programs are linked to the hukou system, which divides citizens into rural and urban categories. Pension schemes are mutually independent across hukou types and assign benefits accordingly. Specifically, China established an old-age security system for urban employees in the early 1990s, featuring relatively high coverage and generous payments (Huang and Zhang 2021). In contrast, rural residents primarily relied on family support until the implementation of the NRPS in 2009. The NRPS targets only rural seniors and operates independently of existing pension policies for urban residents. See Fang and Feng (2018) for a comprehensive overview of China's pension system.

[4]Figure 1 demonstrates the trend of county-level coverage rate, and Appendix Figure A.1 illustrates the county-by-county coverage. Although the implementation of the NRPS could be non-random, our empirical strategy does not rely on the randomization but only requires a common trend.

[5]The central government fully sponsors the basic pension benefit. Although local governments are allowed to provide even more generous pensions, they have to finance extra payments themselves. Thus, the majority of the rural elderly de facto only obtain a basic monthly pension of ¥55 (Huang and Zhang 2021).

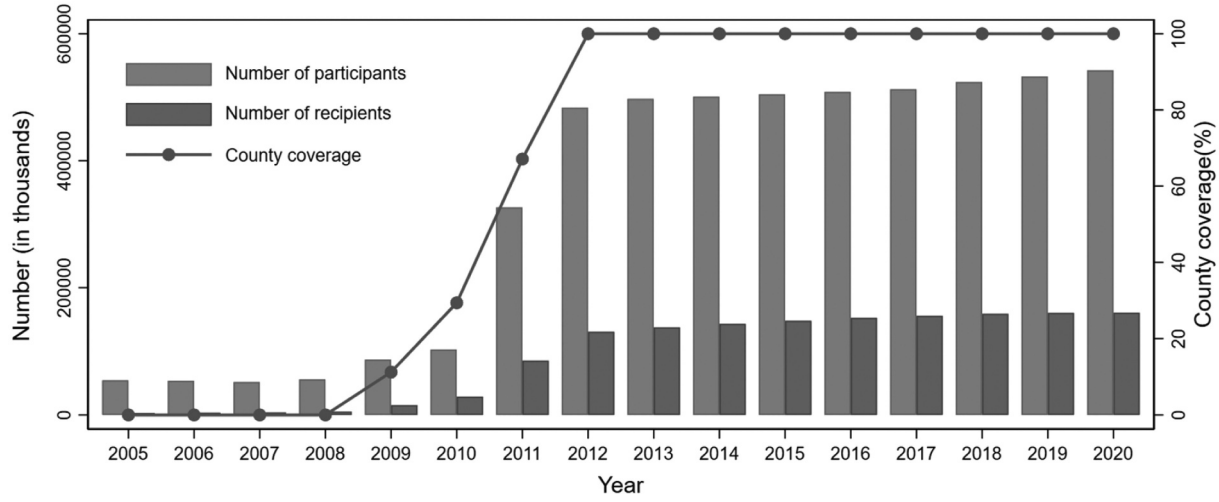


Figure 1. The expansion of the New rural pension Scheme. The data is from the Statistical Bulletin on the Development of Human Resources and Social Security, 2005-2020.

representative sample of Chinese communities, families, and individuals, covering 162 counties. The CHARLS, initiated in 2011, is also a nationally representative biennial survey of families and individuals aged 45 and older, covering 150 counties in China. These two datasets facilitate our analysis in two important ways. First, they provide high-quality microdata with the required information, such as spousal age and the current county of residence, and individuals' age, gender, education, ethnicity, and detailed activity-specific time use. Notably, the ages of both individuals and their spouses are defined using both birth year and month, allowing for more precise eligibility classification. Second, pooling the CFPS and the CHARLS can provide a large nationally representative sample with consistent variables used in this paper (Huang and Zhang 2021).[6] Therefore, we primarily use the data pooling the 2010, 2012, 2014, and 2016 waves of the CFPS and the 2011, 2013, 2015, and 2018 waves of the CHARLS to construct a larger dataset for our empirical analysis.

We make further sample restrictions to facilitate our analysis. First, we primarily focus on age-ineligible individuals who aged at least 45 but strictly less than 60 with spouses age-eligible for pension benefits (aged 60 and above) in rural areas. Second, we limit our analysis to samples with complete information on individuals and their spouses' characteristics. Our final data consists of 4686 rural observations. Moreover, we also use the sample in urban areas with the same selection criteria as a placebo group, which consists of 1362 observations. The sample size for analysis varies with missing patterns of outcome variables.

The primary outcomes and the treatment variable are defined as follows. First, the dependent variables of our primary interest are three variables related to time use: (i) weekly hours spent working, including any work for earnings, such as agricultural or nonagricultural work, agricultural business, and self-employment; (ii) weekly hours spent caring for families, such as the time devoted to caring for elderly parents and grandchildren; and (iii) weekly hours spent exercising, including all vigorous activities requiring high-intensity physical effort that last at least ten minutes at a time, such as walking, running, and bicycling. Second, the treatment variable is an indicator denoting whether the NRPS covers the county.[7] As the pilot counties in each round of expansion were announced in the second half of each calendar year, while the CFPS and the CHARLS surveys were conducted during the summer

[6]We follow Huang and Zhang (2021) and pool the two datasets primarily for three reasons. First, as the CFPS and the CHARLS usually cover different counties, the pooled data has a larger sample size covering more counties so that we can obtain more efficient estimates. Second, the sample can be nationally representative when we include fixed effects for county-by-survey in baseline regressions (Huang and Zhang 2021). Third, the two surveys are conducted by institutes at Peking University and provide consistent variables for analyses.

[7]It is reasonable to use the spousal current county of residence when we lack detailed information on the spousal county of hukou registration as the migration rate among the rural elderly is extremely low (Huang and Zhang 2021).

vacation, the data observed was collected before the implementation of the NRPS in each calendar year. Therefore, we follow Huang and Zhang (2021) and define the treatment variable as whether the county belongs to the previous round of the NRPS pilot. Table 1 reports the summary statistics for the rural sample.[8]

Empirical strategy

We examine the effects of spousal eligibility for pension benefits on individuals' time use in a difference-in-differences (DID) framework. Exploiting the temporal and geographical variation in the NRPS implementation at the county level, we compare individuals (who are age-ineligible themselves but with spouses age-eligible for pension benefits) in the covered counties (treatment group) with those in the uncovered counties (control group) and contrast their differences in time-use patterns after versus before the local enforcement of the NRPS. Specifically, we estimate the following regression equation:

$$TimeUse_{ictps} = \beta_0 + \beta_1 Treat_{ct} + \theta \boldsymbol{X}_{ict} + \lambda_{c,s} + \mu_t + \gamma_p \times t + \varepsilon_{ictps} \quad (1)$$

where the dependent variable $TimeUse_{ictps}$ indicates individual i's weekly hours spent working, caring for families, and exercising in county c province p in year t, drawn from the data source s (the CFPS and the CHARLS). $Treat_{ct}$ denotes whether the county c is affected by the NRPS in year t. $\boldsymbol{X}_{ict}$ refers to covariates, including individuals' demographics (age, gender, years of schooling, and ethnicity), spousal age and education, and the number of children. To tease out potential systematic differences between the two surveys and to account for the differences in counties sampled by the two surveys, we follow Huang and Zhang (2021) and introduce the data source-specific county fixed effects $\lambda_{c,s}$ throughout the analysis. We also include year-fixed effects μ_t to account for time trends in time-use patterns that are common to all counties and control for province-specific linear trends $\gamma_p \times t$ to absorb trends in province-specific factors influencing outcome variables. ε_{ictps} is the error term. All standard errors are clustered at the county level. The coefficient of interest β_1 provides a double-difference estimate of the intention-to-treat (ITT) effect or a lower bound estimate of the effect of spousal social pensions on individuals' time use,[9] and mainly captures the cross-spousal effects of social pensions.[10]

Table 1. Summary statistics of the rural sample.

	Full Sample			Pre-Policy			Post-Policy			
	N	Mean	S. D.	N	Mean	S. D.	N	Mean	S. D.	Difference (Post – Pre)
Panel A: Outcome Variables										
Weekly working hours	4582	10.62	18.62	1371	20.68	21.88	3211	6.32	15.14	−14.36***
Weekly caring hours	2397	28.05	28.06	800	14.64	19.49	1597	34.77	29.27	20.13***
Weekly exercising hours	1944	9.53	9.65	611	1.79	4.04	1333	13.08	9.40	11.30***
Panel B: NRPS Coverage										
Treat = 1	4686	0.70	0.46	1406	0.00	0.00	3280	1.00	0.00	1.000***
Panel C: Control Variables										
Age	4686	57.18	2.76	1406	57.20	2.59	3280	57.20	2.83	−0.03
Female = 1	4686	0.84	0.36	1406	0.86	0.35	3280	0.83	0.37	−0.03**
Years of schooling	4686	2.80	4.00	1406	2.30	3.53	3280	3.01	4.17	0.71***
Ethnic Han = 1	4686	0.92	0.28	1406	0.92	0.27	3280	0.91	0.28	−0.01
Spousal age	4686	62.51	2.72	1406	62.52	2.68	3280	62.50	2.74	−0.02
Spousal years of schooling	4686	4.06	4.13	1406	3.99	3.86	3280	4.10	4.24	0.10
Number of children	4686	2.01	1.30	1406	2.05	1.31	3280	2.00	1.30	−0.05

We focus on individuals who themselves are age-ineligible (aged at least 45 but strictly less than 60) with spouses age-eligible (60 years or older) for pension receipt. Data is from the CFPS (waves 2010, 2012, 2014, and 2016) and the CHARLS (waves 2011, 2013, 2015, and 2018).

[8]Appendix Table A.1 also shows summary statistics for the urban sample.
[9]As we focus on all individuals with spouses age-eligible for pension benefits in covered counties, this coefficient captures the ITT effect.
[10]Given that we restrict to individuals who themselves are not age-eligible to pension benefits, β_1 reflects the cross-spousal effects.

The key identifying assumption of the double-difference estimation strategy is a parallel trend in outcome variables. It requires that individuals in covered counties and those in unaffected counties would have followed the same trends in time-use patterns in the absence of the social pension programme. Although we cannot stringently examine the parallel pre-trends using an Event Study approach due to data limitations,[11] we provide suggestive evidence for the assumption. Specifically, we follow Huang and Zhang (2021) and plot a set of county-level characteristics over calendar years by the round of the NRPS. These characteristics could simultaneously affect individuals' time use and the local enforcement of the NRPS. Panel A of Appendix Figure A.2 illustrates that the time trends for GDP per capita are fairly parallel between counties that implemented the NRPS in different rounds. Similar patterns are observed for all other county-level characteristics, including savings, government expenditure per capita, and government revenue per capita. We also conduct F-tests for parallel trends in these characteristics. The F-statistics and corresponding p-values are reported in each panel. These tests indicate no significant nonparallel trends between counties with different NRPS starting years.

IV. Empirical results

Effects on pension receipt

Before delving into the effects of spousal social pensions on individuals' time use, we first evaluate the policy's effectiveness by examining its impact on pension receipt. Figure 2 illustrates the age-specific coefficient using Equation (1) for the rural and urban sample, respectively. The figure shows minimal effects on urban people across all age groups and rural people younger than 60 but a significant positive impact on rural people aged 60 years and above, which bolsters the policy's effectiveness and our sample selections.[12]

Cross-spousal effects on time use

Table 2 reports our estimates on the impact of spousal eligibility for pension benefits on individuals' time use. Columns (1) to (3) present the estimation results controlling for baseline fixed effects for year and county-by-survey. Columns (4) to (6) further control for a set of control variables, including individuals' characteristics (age, gender, years of schooling, and ethnicity), spousal characteristics of age and education, and the number of children. Columns (7) to (9) further introduce province-specific linear trends. Panel A shows significant and robust declines in weekly working hours and increases in weekly hours spent caring for families and exercising for the rural sample. At the same time, Panel B shows no significant effects on the urban sample that is ineligible for participating in the NRPS, which aligns with existing findings (e.g. Huang and Zhang 2021).[13]

Our preferred specification in columns (7) to (9) in Panel A of Table 2 shows that spousal eligibility for pension benefits from the NRPS significantly decreased individuals' weekly time spent working by approximately 5.3 hours, on average, while substantially increasing individuals' time spent caring for families and exercising by approximately 7.5 hours and 4.1 hours each week, respectively. Given the sample mean of outcome variables, spousal eligibility for pension benefits decreased individuals' time spent working, caring, and exercising by 49.5%, 26.7%, and 42.6%, respectively. Our findings on individuals' decreased working hours and increased time spent caring are consistent with the effects of social pensions on the beneficiaries themselves (Hua, Zhang, and Liu 2022; Shu 2018) but with slightly smaller estimates.[14] This can be plausibly attributed to two reasons. First, the benefits from the NRPS, ¥55 (approximately US $9) per month, may be less generous than those from other social pension programmes. For example, a non-contributory pension scheme in Mexico initiated in 2007 provided 500 Mexican pesos (approximately US $40) per month for the

[11]We have a limited sample in pre-periods to examine the parallel trend using an event study approach.
[12]Appendix Table A.2 reports estimates for subsamples by hukou type and age groups.
[13]We also employ a triple-difference strategy by interacting the treatment variable with an indicator for rural hukou to provide additional evidence. Appendix Table A.3 also reports significant estimates, which further shows that the NRPS's impact is concentrated among rural residents.
[14]For example, we find a 5.3 reduction in individuals' weekly working hours, which is slightly more moderate than a decline of 22.5 hours induced by a Brazilian social pension reform (Carvalho Filho and Evangelista 2008).

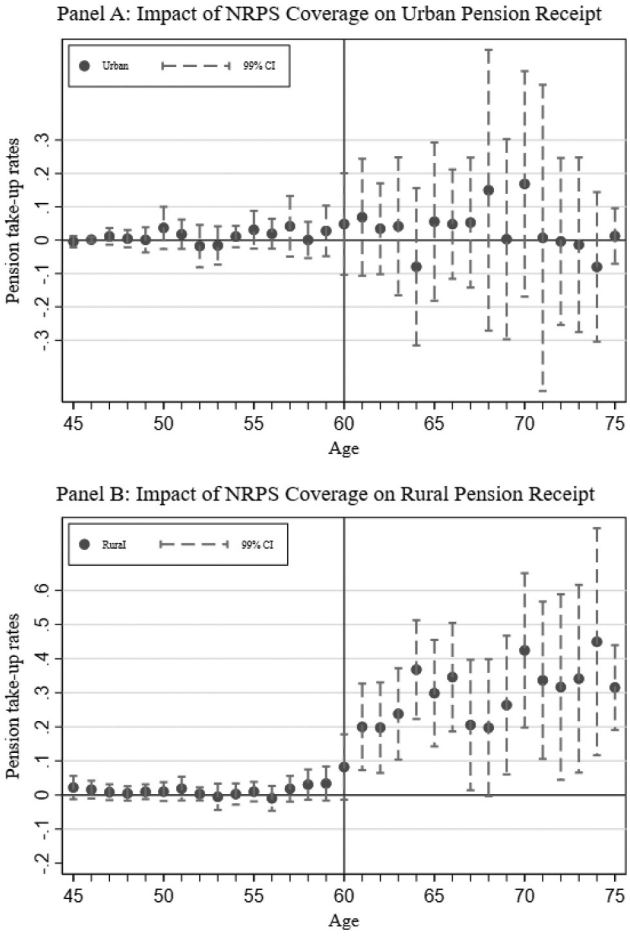


Figure 2. Effects on pension receipt by hukou type across age Groups. This figure illustrates the age-specific coefficients with pension receipt as the dependent variable using regression specification in Equation (1) for urban and rural groups, respectively. The dashed lines indicate the 99% confidence interval.

rural elderly (Ávila-Parra, Escamilla-Guerrero, and Gálvez-Soriano 2024). Second, the household income from spousal pension benefits may not be shared effectively within a couple (Juarez and Pfutze 2015).

Heterogeneous cross-spousal effects

This section examines the heterogeneity of the cross-spousal effects by local economic development and individuals' gender. Specifically, we first use a lower county-level GDP per capita and whether located in the mid-western region to proxy for a lower local economic development level. We then re-estimate the results for women and men subsamples, respectively. Table 3 consistently shows that the NRPS had more pronounced effects on disadvantaged people, particularly women and those from less-developed regions, which aligns with existing findings (Lundberg 1985; Shu 2018).

Robustness checks

Appendix Section B provides details for all robustness checks as follows.

Potential confounders

This subsection addresses concerns from several potential confounders. First, we incorporate county-specific linear trends and individual fixed effects to absorb unobserved factors at the county or individual level. Appendix Table B.1 presents robust estimates. Second, we further show that our results are not biased by selective migration behaviours as Appendix Table B.2 shows that the NRPS had minimal effects on the family size and cross-county migration. Third, we address the anticipation effect of own eligibility by controlling for year-specific linear age trends and by using a younger sample (e.g. aged 55 and below) instead. Appendix Table B.3 reports robust estimates. Finally, we control for two concurrent policies: China's Universal Two-Child Policy, which might increase the demand for grandchild care (Eibich and Siedler 2020), and Three Rights Separation Land Reform, which can increase rural household income (Peng, Yang, and Chen 2020). Appendix Table B.4 again presents robust results.

Bias in staggered DID

To account for potential bias in the staggered DID approach, we follow the intuition outlined in Goodman-Bacon (2021) and adopt a stacked DID approach to restrict our comparisons to 'switcher' units and 'not-yet-treated' units. Appendix Table B.5 presents robust results.

NRPS as instrumental variable

We use the NRPS as an instrument variable for spousal participation in the programme to estimate the local average treatment effect (LATE). Appendix Table B.6 shows that spousal participation in the NRPS influenced individuals' time-use outcomes more significantly.

Table 2. Cross-spousal effects on individuals' time use.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours
Treat = 1	−5.252**	7.496**	4.056**	−5.013**	7.198**	4.092***	−5.474**	7.856**	3.357**
	(2.256)	(3.496)	(1.669)	(2.246)	(3.491)	(1.607)	(2.458)	(3.769)	(1.691)
Dependent Mean	10.617	28.050	9.531	10.617	28.050	9.531	10.617	28.050	9.531
Observations	4,582	2,397	1,944	4,582	2,397	1,944	4,582	2,397	1,944
R^2	0.337	0.326	0.512	0.351	0.340	0.524	0.362	0.353	0.536
Treat = 1	4.130	−1.563	3.495	4.284	0.535	3.483	5.888	5.911	4.724
	(3.881)	(7.318)	(3.534)	(4.161)	(6.980)	(3.328)	(5.029)	(5.835)	(2.954)
Dependent Mean	4.033	22.678	7.475	4.033	22.678	7.475	4.033	22.678	7.475
Observations	1,321	448	606	1,321	448	606	1,321	448	606
R^2	0.256	0.459	0.406	0.276	0.480	0.428	0.315	0.525	0.469
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
County-by-Survey FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Control Variables				✓	✓	✓	✓	✓	✓
Province-Specific Linear Trends							✓	✓	✓

Control variables include individuals' characteristics (age, gender, years of schooling, and ethnicity), spousal characteristics of age and education, and the number of children. Standard errors in parentheses are clustered at the county level. * Significance at the 10 percent level. ** Significance at the 5 percent level. *** Significance at the 1 percent level.

Table 3. Heterogeneous cross-spousal effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours
		High Level of GDP			*Low Level of GDP*	
Treat = 1	−3.429	7.109	0.294	−6.648**	8.298*	5.733***
	(3.856)	(5.922)	(2.772)	(3.271)	(4.831)	(2.149)
Observations	2,098	1,028	919	2,484	1,208	1,025
R^2	0.331	0.339	0.507	0.395	0.376	0.577
		Eastern Region			*Mid-Western Region*	
Treat = 1	−6.504	8.228	1.319	−4.215	8.108*	5.049**
	(4.195)	(6.430)	(2.857)	(3.169)	(4.426)	(2.139)
Observations	1,627	736	687	2,955	1,661	1,257
R^2	0.318	0.363	0.532	0.389	0.350	0.547
		Male			*Female*	
Treat = 1	−11.131	21.487	6.635	−5.841**	9.785**	3.475*
	(9.615)	(20.909)	(7.299)	(2.605)	(4.196)	(1.989)
Observations	594	245	204	3,873	2,044	1,646
R^2	0.540	0.516	0.739	0.394	0.376	0.580
Year FE	✓	✓	✓	✓	✓	✓
County-by-Survey FE	✓	✓	✓	✓	✓	✓
Control Variables	✓	✓	✓	✓	✓	✓
Province-Specific Linear Trends	✓	✓	✓	✓	✓	✓

This paper refers to the Statistical Communiqué of the People's Republic of China National Economic and Social Development to divide the sample into eastern and central-western regions. Control variables include individuals' characteristics (age, gender, years of schooling, and ethnicity), spousal characteristics of age and education, and the number of children. Standard errors in parentheses are clustered at the county level. * Significance at the 10 percent level. ** Significance at the 5 percent level. *** Significance at the 1 percent level.

Data issues

We further address three data issues. First, we improve the data's national representativeness by reweighting the sample and using survey data separately. Appendix Table B.7 shows robust estimates. Second, we mitigate the concern from different missing patterns of three outcome variables by showing that the NRPS had minimal impact on whether the outcome is missing in Appendix Table B.8. Third, we alleviate the concern from the time gap between the NRPS implementation and the post-reform periods by using the data of 2010–2015 or 2010–2016 instead. Appendix Table B.9 reports robust estimates.

Placebo tests

We further conduct two placebo tests. First, we re-estimate the effects using a falsified minimum eligibility age and a placebo sample in which individuals' spouses are actually age-ineligible. Appendix Table B.10 consistently reports minimal impact

when we falsify the age at 59, 57, 55, and 53. Second, we randomly assign the treatment variable 1000 times, and Appendix Figure B.1 consistently shows null effects.

V. Mechanism analysis

Income effect

The first plausible mechanism leading to less work and more leisure for individuals is the income effect of spousal pension benefits. Governments provided a basic pension benefit of ¥55 (approximately US $9) per month for rural people aged 60 years and older without paying any premium immediately after the local implementation of the NRPS. This basic pension benefit is sufficient to guarantee the basic survival of an older person in rural China (Huang and Zhang 2021). Moreover, given that the median monthly income per capita in households with members aged 60 or older is only ¥302, and there are approximately 5.05 percent of rural elderly whose household income per month is less than ¥55 in our analysis sample, the amount of the basic pension benefit is obviously nontrivial. Hence, spousal pension benefits, or a windfall cash transfer, could affect individuals' time allocation through the income effect (Gale 1998; Huang and Zhang 2021). We would thus expect to observe more pronounced effects among individuals from low-income households.

To examine this mechanism, we first explore the impact of spousal eligibility for pension benefits on household income. As reported in Columns (1) and (2) of Panel A in Table 4, the NRPS significantly increased household income and per capita income in households with members who were age-eligible for pension benefits (aged 60 and above). It is noteworthy that the effects on household income may have been attenuated by the fact that the recipients and their family members may have reduced their labour supply significantly. Therefore, our findings of significant increases in household income suggest that the cash transfers from spousal pension benefits are substantial enough even after offsetting the forgone earnings from less work, which is consistent with the literature (Huang and Zhang 2021).

We further examine the income effect by exploring heterogeneity based on household income levels. Specifically, we estimate the impact of the NRPS on individuals' time use outcomes separately for households with higher and lower per capita income. The estimated results, presented in Columns (3)-(5) and (6)-(8) of Panel A in Table 4, respectively, consistently show that the NRPS had a significant effect on individuals from low-income households, while having minimal

Table 4. Mechanism analysis.

	(1)	(2)			(3)	(4)	(5)	(6)	(7)	(8)
	Full Sample				Higher Household Income per capita			Lower Household Income per capita		
	Ln (Household Income)	Ln(Household Income per capita)			Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Working Hours
Treat = 1	0.132**	0.729**			−0.342	3.356	5.513	−7.707***	16.240**	4.540**
	(0.064)	(0.294)			(6.640)	(5.246)	(5.169)	(2.742)	(6.435)	(2.022)
Observations	4,686	4179			1927	895	627	1991	902	662
R^2	0.469	0.437			0.417	0.376	0.563	0.460	0.377	0.529
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Full Sample				Lower Baseline County-level Pessimism Index			Higher Baseline County-level Pessimism Index		
	Hopeless	Meaningless	Dissatisfied	Pessimism Index	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours	Weekly Working Hours	Weekly Caring Hours	Weekly Exercising Hours
Treat = 1	−0.162**	−0.116***	−0.106**	−0.481***	0.131	14.270	−2.277	−8.884**	11.864*	9.174**
	(0.068)	(0.044)	(0.050)	(0.154)	(6.612)	(9.233)	(1.636)	(3.507)	(6.309)	(3.648)
Observations	4016	4449	3826	3372	1939	841	656	1909	835	618
R^2	0.327	0.214	0.177	0.232	0.429	0.574	0.661	0.453	0.404	0.643

Lower Household Income per capita indicates the household's income per capita is lower than the sample median. Pessimism Index is a standardized index of pessimism using principal component analysis (PCA). Higher Baseline County-level Pessimism Index is a binary indicator equal to one if the county-level average of the pessimism index, measured when the county was first observed in our main sample, is above the median. All regressions have controlled for year fixed effects, county-by-survey year fixed effects, control variables, and province-specific linear trends. Control variables include individuals' characteristics (age, gender, years of schooling, and ethnicity), spousal characteristics of age and education, and the number of children. Standard errors in parentheses are clustered at the county level. * Significance at the 10 percent level. ** Significance at the 5 percent level. *** Significance at the 1 percent level.

impact on those from higher-income households. These findings provide additional support for the income effect channel.

Expectation effect

The second mechanism might be that spousal pension benefits can also affect individuals' time use by forming an optimistic expectation. A monthly basic pension benefit of ¥55 is similar to a defined-benefit plan, which can provide transparent and predictable cash inflow to households in the future. Predictable household income might reduce individuals' feelings of uncertainty and encourage them to be more optimistic about the future (Xu et al. 2022). Forward-looking individuals make time-use decisions considering their life-cycle income, and beliefs about the future could also affect current behaviours. Hence, the transparent and predictable cash inflow from spousal pension benefits in the future could affect individuals' current time-use patterns through the expectation effect. Additionally, this effect is likely to be more pronounced among individuals residing in areas where people tend to be more pessimistic.

To examine this mechanism, we first investigate the impact of the NRPS on individuals' attitudes towards future life. Specifically, we use three binary indicators to measure individuals' pessimism: (i) feeling hopeless towards the future, (ii) feeling that life is meaningless, and (iii) feeling dissatisfied with life. Each variable equals one if the respondent experienced such feelings in the past week, and zero otherwise. In addition, we construct a standardized index of pessimism using principal component analysis (PCA). As shown in Columns (1) to (4) of Panel B in Table 4, spousal pension benefits significantly reduced individuals' feelings of hopelessness, meaninglessness, and dissatisfaction with life. These findings suggest that the NRPS helps foster a more optimistic outlook towards the future.

Next, we assess whether the policy-induced optimism translated into changes in individuals' time use. To do so, we analyse the heterogeneity of the policy's impact based on baseline levels of pessimism at the county level. Specifically, we construct a binary indicator equal to one if the county-level average of the pessimism index, measured when the county was first observed in our main sample, is above the median. We then separately estimate the NRPS's effects on individuals from counties with higher versus lower baseline pessimism. The results, presented in Columns (5)-(7) and (8)-(10) of Panel B in Table 4, consistently indicate that the NRPS had a stronger impact on time use outcomes for individuals in more pessimistic areas. These findings provide further evidence that the NRPS can influence time allocation by promoting optimistic expectations about the future.

VI. Further analysis

Cross-spousal effects on the relationship with non-cohabitant children

To provide additional evidence on the significant influence of spousal social pensions, this section investigates the cross-spousal spillover effects of the NRPS on individuals' relationships with their non-cohabitant children. The underlying rationale is that public transfers may crowd out private transfers and potentially improve elderly parents' emotional closeness with their adult children (Bando, Galiani, and Gertler 2022; Jensen 2004). If pension benefits received by one spouse are shared within the couple, they may also influence the other spouse's receipt of private transfers from their children and alter the quality of intergenerational relationships. Prior to the universal coverage of the NRPS, rural elderly individuals primarily relied on family-based economic support (Kaushal 2014), especially private transfers from their adult children. This financial dependency may have adversely affected emotional intimacy between parents and children. However, public pension programmes have been widely shown to reduce such financial dependence, often crowding out private transfers from working-age adults (Jensen 2004), influencing living arrangements (Bando, Galiani, and Gertler 2022), and even reshaping traditional kinship practices (Bau 2021). Despite this, little is known about whether social pensions received by one spouse also reduce private transfers received by the other spouse and

Table 5. Further analysis.

	(1)	(2)	(3)	(4)
	Private Transfer from Children	Contacts Frequency	Physical Unhealthiness Index	Mental Unhealthiness Index
Treat = 1	−607.820***	0.126**	−0.124	−0.313**
	(209.821)	(0.060)	(0.153)	(0.159)
Observations	1,857	1,990	3,321	3,403
R^2	0.258	0.231	0.244	0.300
Year FE	✓	✓	✓	✓
County-by-Survey FE	✓	✓	✓	✓
Control Variables	✓	✓	✓	✓
Province-Specific Linear Trends	✓	✓	✓	✓

Control variables include individuals' characteristics (age, gender, years of schooling, and ethnicity), spousal characteristics of age and education, and the number of children. Standard errors in parentheses are clustered at the county level. * Significance at the 10 percent level. ** Significance at the 5 percent level. *** Significance at the 1 percent level.

whether such changes improve intergenerational relationships.

To provide empirical evidence, we explore the impact of spousal eligibility for pension benefits on individuals' cash transfers from their adult children and their contact frequency with non-cohabitant children, respectively. First, Column (1) in Table 5 shows that spousal eligibility for the NRPS significantly decreased individuals' private transfers from adult children, which is seemingly different from the findings of Huang and Zhang (2021), who found minimal impact. There are two plausible reasons accounting for this difference: (i) we focus on rural individuals who are age-ineligible themselves with spouses age-eligible for pension benefits instead of all rural elderly, and (ii) we focus on individuals' private transfer only from children instead of from all sources. Second, Column (2) in Table 5 presents that spousal social pensions can also significantly increase individuals' frequencies of contact with their non-cohabitant children. Together, these findings suggest that spousal social pensions reduced individuals' financial dependence on children, while children may respond by offering more non-monetary support such as increased contact and emotional connection.

Cross-spousal effects on health outcomes

We also examine the cross-spousal spillover effects of the NRPS on individuals' physical and mental health – an especially relevant extension, given our findings on time-use changes and the growing attention to elderly health outcomes in the literature (Cheng et al. 2018; Miglino et al. 2023). Specifically, the existing literature shows that social pensions could benefit the physical health and cognitive function of the rural elderly (Huang and Zhang 2021; Miglino et al. 2023) through reduced malnutrition, better access to healthcare, and increased leisure activities (Cheng et al. 2018). Although we have documented a significant and robust increase in time spent exercising due to spousal eligibility for pension benefits, we currently know little about how individuals' health performance is influenced.

First, we explore the impact of spousal eligibility for pension benefits on individuals' physical health. To begin with, we construct an index representing physical unhealthiness. We first define four indicators of unhealthiness: (i) abnormal weight, which takes the value of one if an individual is malnourished (BMI <18.5) or obese (BMI >25) and values zero otherwise; (ii) inpatient healthcare utilization, which takes the value of one if an individual had inpatient visits in the last year and values zero otherwise; (iii) self-reported unhealthiness, which takes the value of one if an individual report physically unhealthy and values zero otherwise; and (iv) self-reported disability, which takes the value of one if an individual reports demand for helps in daily life activities, such as walking, cooking, dining, travelling, shopping, and doing housework, and values zero otherwise. Then, we use principal component analysis (PCA) to obtain a z-scored index for physical unhealthiness. The results in Column (3) in Table 5 suggest that the NRPS had minimal importance in improving individuals' physical health, which is consistent with existing findings (Bando, Galiani, and Gertler 2022).

Second, we also investigate the cross-spousal impact on individuals' mental health. Similarly, we first construct an index denoting mental unhealthiness. According to the depression scale, we define six indicators of depression: nervousness, restlessness, despair, difficulty, and meaninglessness, respectively. All indicators take the value of one if individuals report that they have the corresponding syndrome sometimes and value zero otherwise. Then, we use principal component analysis (PCA) and get a z-scored index for mental unhealthiness. In contrast to trivial effects on physical health, Column (4) in Table 5 shows that the NRPS significantly decreased mental unhealthiness by 0.31 standard deviations. These results indicate that social pensions can not only improve beneficiaries' but also benefit their spouses' mental health (Bando, Galiani, and Gertler 2022; Cheng et al. 2018). Moreover, the substantial improvement in mental health can be plausibly attributed to three factors: (i) changes in time composition towards more enjoyable activities, i.e. less work (Bando, Galiani, and Gertler 2022) and more leisure activities (Knabe et al. 2010); (ii) increased income and more optimistic expectations about the future; (iii) improved relationship with non-cohabitant children.

VII. Conclusion

This paper investigates the cross-spousal spillover effects of social pensions on individuals' time use. Drawing on two large nationally representative datasets, we exploit the temporal and geographical variation in the rollout of China's New Rural Pension Scheme (NRPS) in a difference-in-differences (DID) framework to identify the cross-spousal effects. We find that spousal eligibility for pension benefits significantly affected individuals' time use, leading to less time spent working and more time spent caring for families and exercising, which were accrued to women and those from less-developed regions. These results are robust to potential confounders, bias in staggered DID, data issues of representativeness and missing patterns, and placebo tests. Mechanism analysis shows that spousal social pensions can affect individuals' time use through the income effect of the pension benefit and the optimistic expectation effect of the transparent and predictable cash inflow from spousal pension benefits. Further investigation indicates that spousal eligibility for pension benefits can also improve individuals' relationships with non-cohabitant children by alleviating the financial burden on children and significantly enhance individuals' mental health, while having minimal impact on physical health.

Our findings have several important implications. First, the significant cross-spousal spillover effects observed in our study indicate that the benefits of social pension programmes extend beyond the direct recipients to other household members, particularly spouses. Neglecting these indirect effects may lead to an underestimation of the overall social returns from such programmes. Recognizing and incorporating these broader benefits could incentivize governments in developing economies to adopt or expand pension schemes. Second, our heterogeneity analysis reveals that the effects are substantially stronger among women and individuals residing in less-developed regions. This suggests that even with limited resources, developing countries can achieve considerable welfare gains by effectively targeting vulnerable populations, especially women and residents of disadvantaged areas. Third, our mechanism analysis underscores the importance of both income and expectation effects stemming from the receipt of transparent and predictable pension benefits by spouses. Accordingly, to enhance the efficacy of pension programmes, governments, particularly in developing countries, should ensure the financial sustainability of pension funds and improve the transparency and predictability of benefits. Fourth, although rural social pensions primarily aim to provide economic support to elderly individuals, we also find that these programmes can improve relationships with non-cohabitant children and enhance the mental health of spouses. This indicates that for policymakers in developing countries, pension schemes can also serve broader goals related to intergenerational support and subjective well-being.

While we have documented insights on the cross-spousal impact of the NRPS on individuals' own time use outcomes, future research can investigate other important questions. For example, to provide further evidence on the cross-spousal

spillover effects of the NRPS, we examine the impact of spousal eligibility for pension benefits on individuals' physical and mental health. Our results reveal minimal impact on physical health but significantly positive effects on mental health. These findings suggest that it is worthwhile to further investigate the impact of the NRPS on mental health, particularly among women and individuals from less-developed regions.

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Data availability statement

The data that support the findings of this study are available at https://opendata.pku.edu.cn/.

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